

Software Project Lab – 1 Proposal Presentation

Project Title: ImCloneDetect (Image-Based Code Clone Detection Tool)

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Introduction

What is code clone?

```
public static void BubbleSortLong1 ( long[] num ) {  
    boolean flag = true;  
    long temp;  
    while ( flag ) {  
        flag = false;  
        for ( int j = 0; j < num.length - 1; j++ ) {  
            if ( num[j] > num[j + 1] ) {  
                temp = num[j];  
                num[j] = num[j + 1];  
                num[j + 1] = temp;  
                flag = true;  
            }  
        }  
    }  
}
```

(a) The first method, BubbleSortLong1

[2]

```
public static void BubbleSortFloat2 ( float[] num ) {  
    int last_exchange;  
    int right_border = num.length - 1;  
    do {  
        last_exchange = 0;  
        for ( int j = 0; j < num.length - 1; j++ ) {  
            if ( num[j] > num[j + 1] ) {  
                float temp = num[j];  
                num[j] = num[j + 1];  
                num[j + 1] = temp;  
                last_exchange = j;  
            }  
        }  
        right_border = last_exchange;  
    } while ( right_border > 0 );  
}
```

(b) The second method, BubbleSortFloat2

[2]

Project Description – “What?”

Tool Development

Develop an Image-Based Code Clone Detection Tool designed for precision and efficiency.

Project Description – “How?”

Core Algorithms

Utilize Jaccard Distance [1], Perceptual Hash (pHash) [1], and Earth Mover's Distance (EMD) [2].

Image Similarity

Detect similarities among source code images (PNG files) using advanced algorithms.

Output & Visualization

Generate similarity results in CSV files and display them in the terminal, enabling visualization in future.

Project Scope

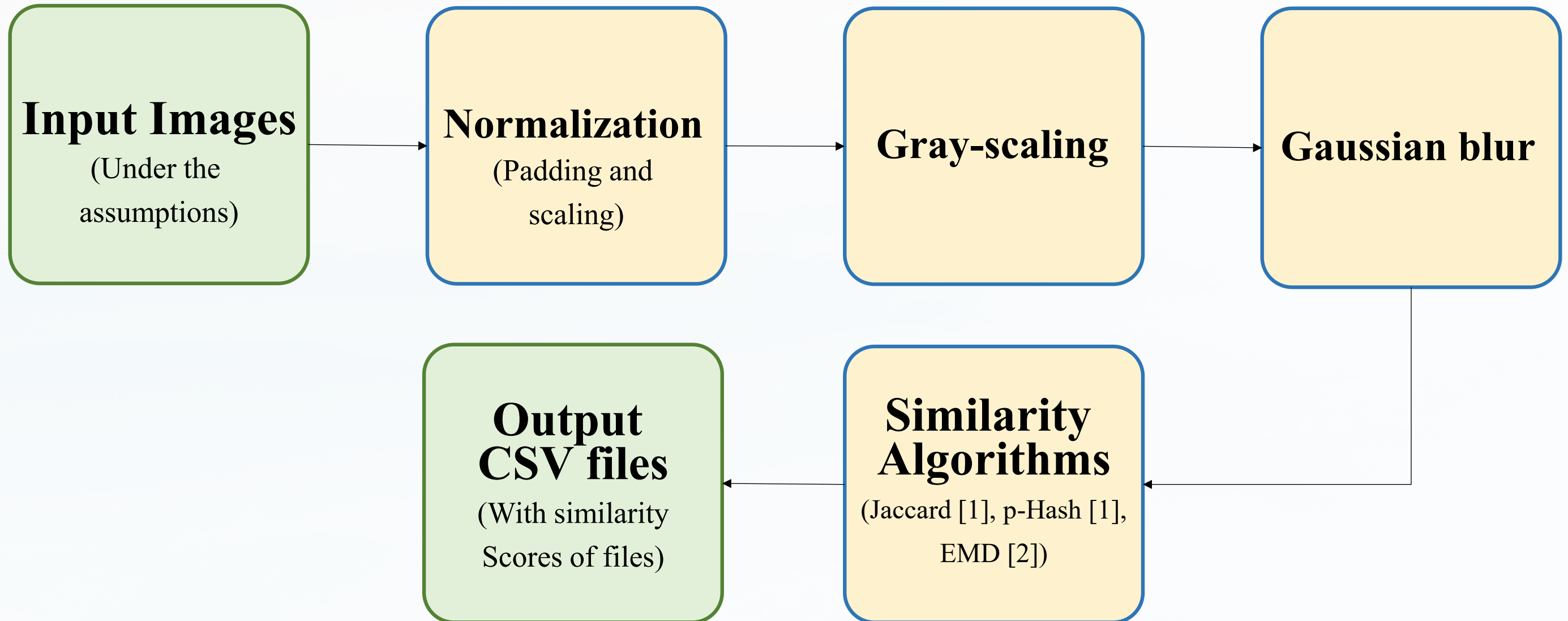
What I Will Do:

- Implement CLI-based image similarity system
- Apply three independent algorithms (Jaccard [1], pHash [1], EMD [2])
- Store and display results effectively
- Prepare CSV outputs for external visualization

Assumptions:

- The input images contain codes without whitespaces and comments
- Images must be functions of same language
- Codes in the images must be syntax highlighted

Work Flow



Technology Stack

Programming Language

C++

IDE

Codeblocks, VS Code

Version Control

GitHub

References

[1] Y. Wang and D. Liu, “Image-Based Clone Code Detection and Visualization,” 978-1-7281-4691-1/19/\$31.00 ©2019 IEEE DOI 10.1109/AIAM48774.2019.00041, pp. 168–175, Oct. 2019, doi: 10.1109/aiam48774.2019.00041.

[2] C. Ragkhitwetsagul, J. Krinke, and B. Marnette, “A picture is worth a thousand words: Code clone detection based on image similarity,” 978-1-5386-6430-8/18 © 2018 IEEE IWSC 2018, Campobasso, Italy. Accepted 10.1109/iwsc.2018.8327318. Accepted 10.1109/iwsc.2018.8327318. for Publication by IEEE, Mar. 2018, doi:

Thank You & Q&A

Thank You for your attention!

Questions and Answers